



Peak Memory

Kumar Abhinandan

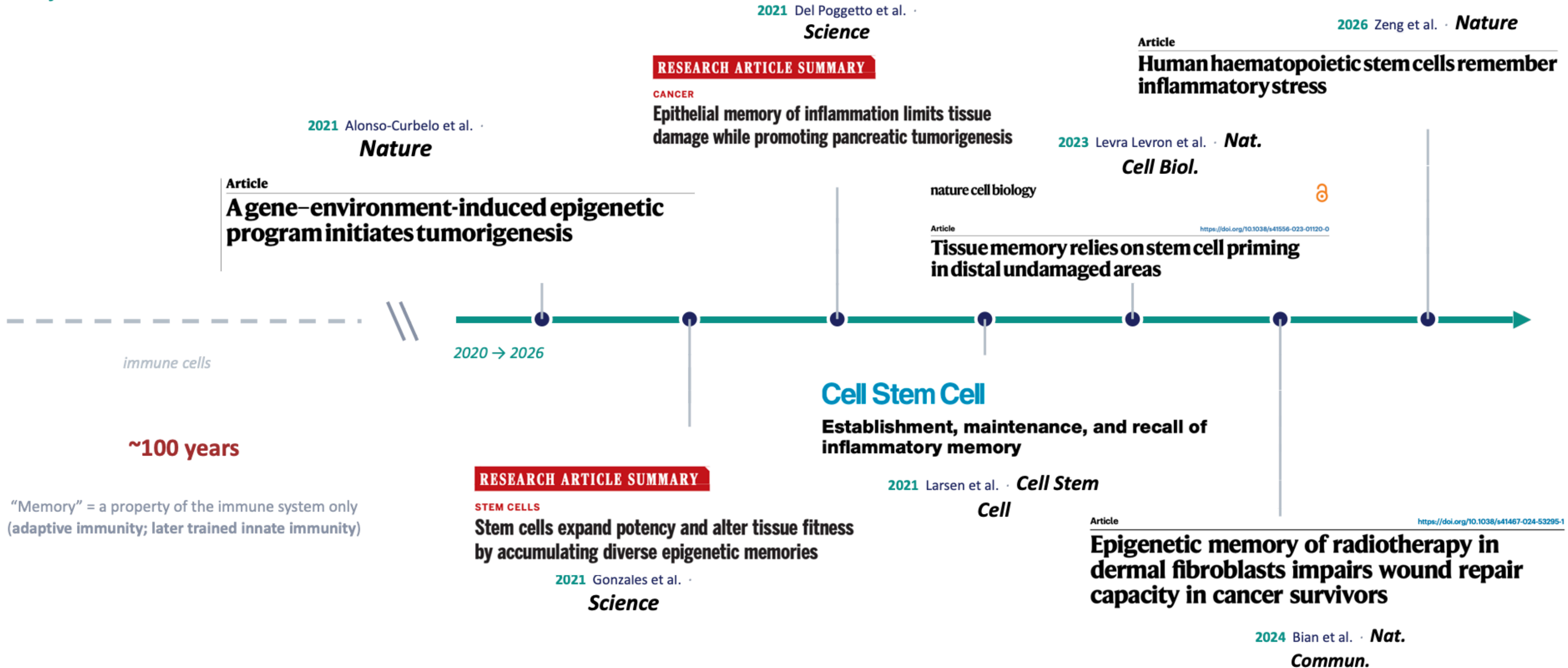
Chandrasekaran

Ayush Munshi

Sampoorn Pandey

Inflammatory Memory: from immune cells to stem cells

For ~a century, immunological memory was thought to belong only to immune cells. **In the last ~5 years, a wave of studies showed that epithelial & tissue stem cells also remember inflammation.**

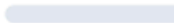


7 studies, 2020–2026 · inflammatory / epigenetic memory now established across epithelial, fibroblast & hematopoietic stem cells

Memory Peaks

How a memory peak is defined

 open region (ATAC peak)

 closed (no peak)

Day 0 (baseline / control)

Injury day 1

Injury day 7

Memory (chase)



union →
memory peak

not open in memory

open at baseline
→ removed

✓ **SELECTED**

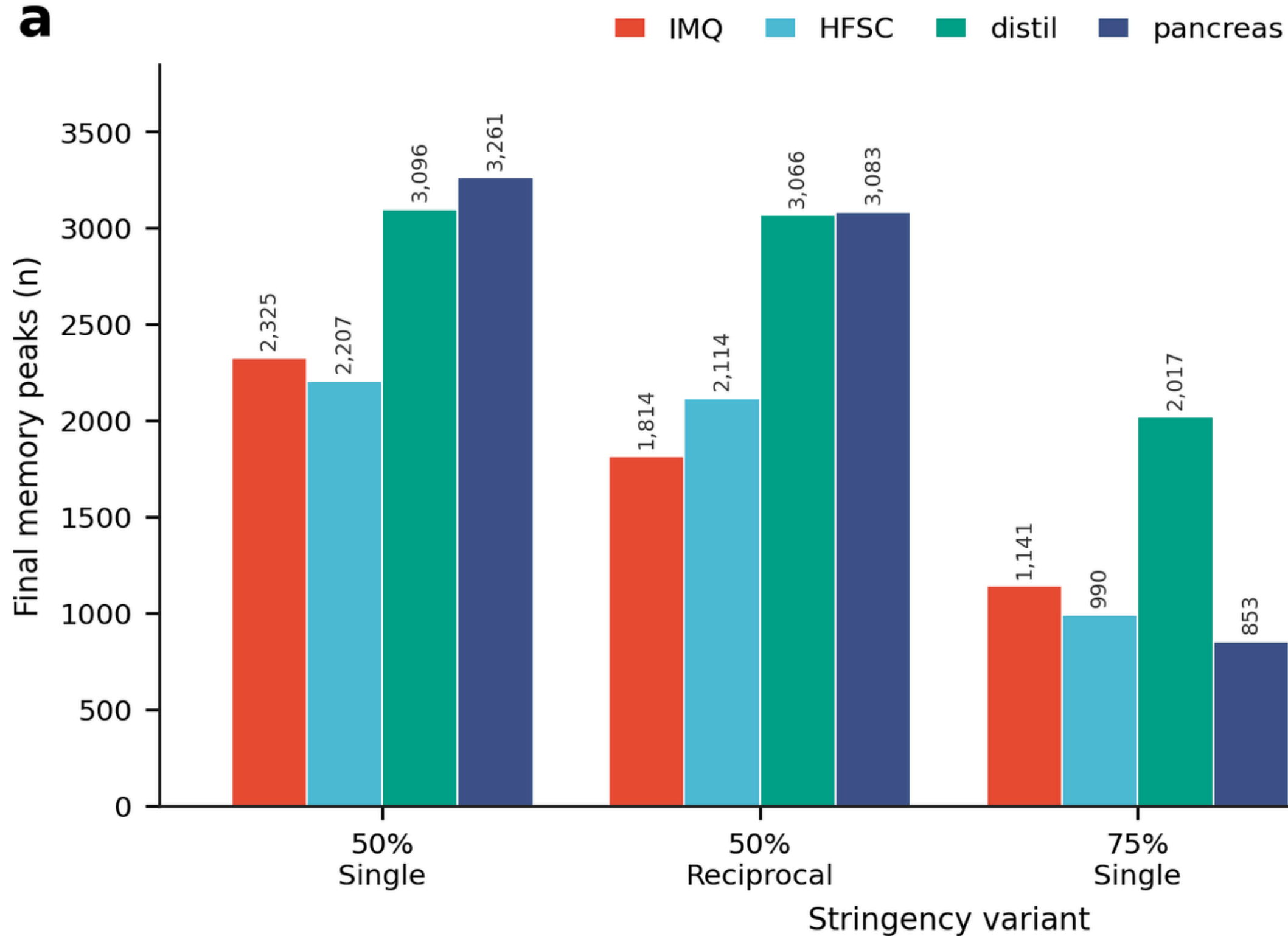
✗ **not retained**

✗ **removed**

Rule: open across injury days (consensus) + still open in memory (chase) + absent at baseline = memory peak

Memory Peaks

Acquired memory peaks per dataset (strict, control-subtracted)

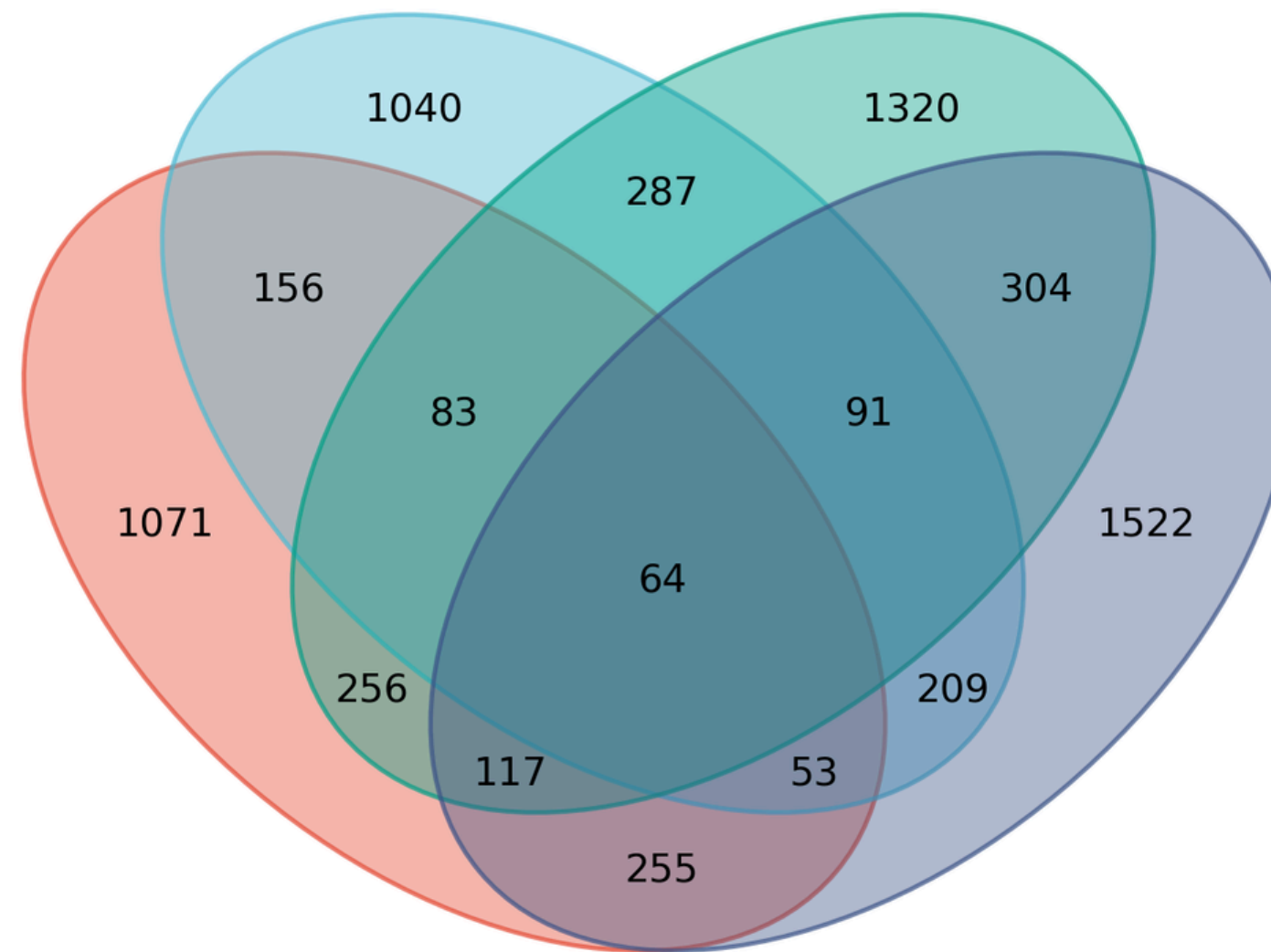


**Non Significant
Peak Overlap**

Annotated Genes

Memory-gene overlap across four datasets (50_Single)

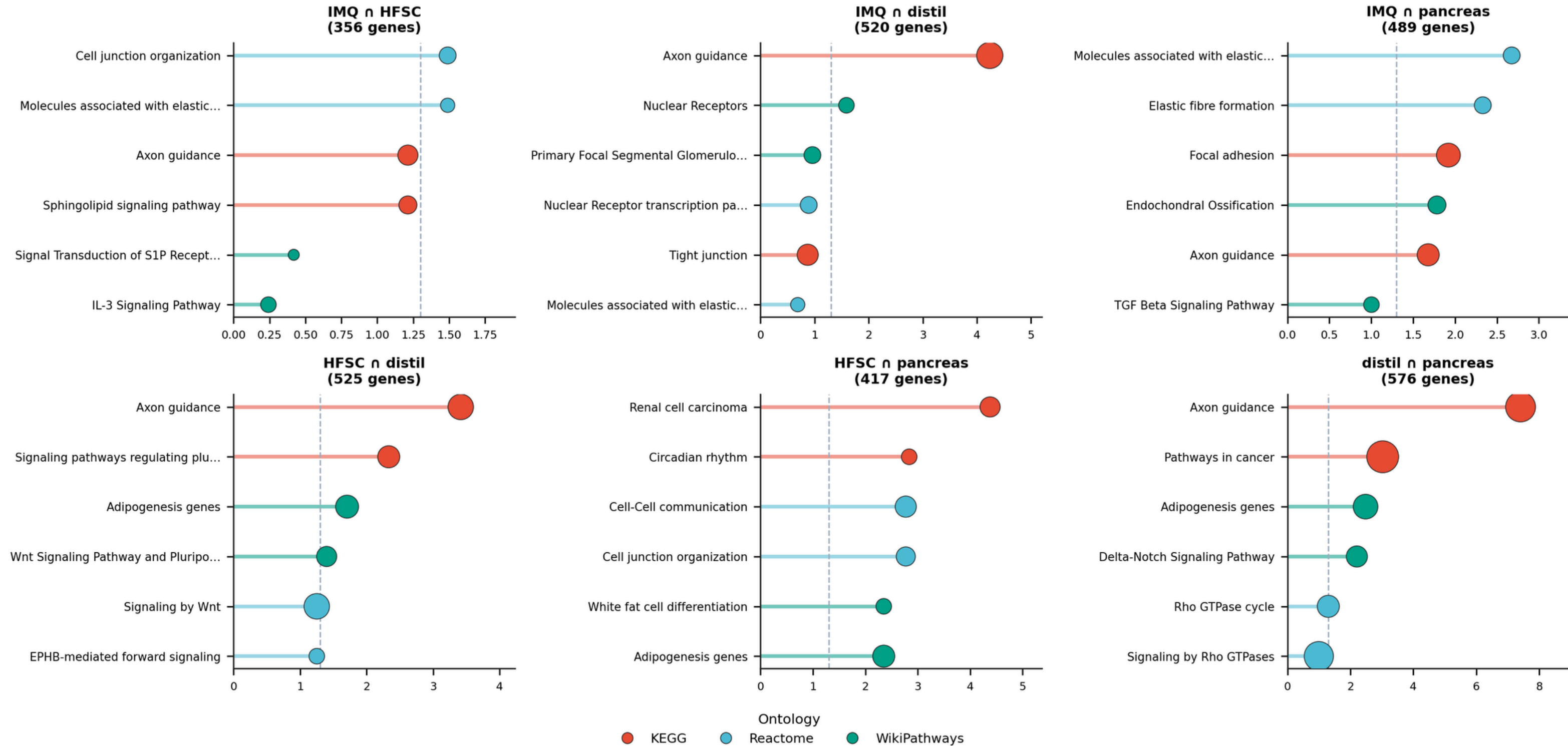
IMQ
HFSC
distil
pancreas



four-way core = 64 genes

Pathway enrichment for Annotated Genes

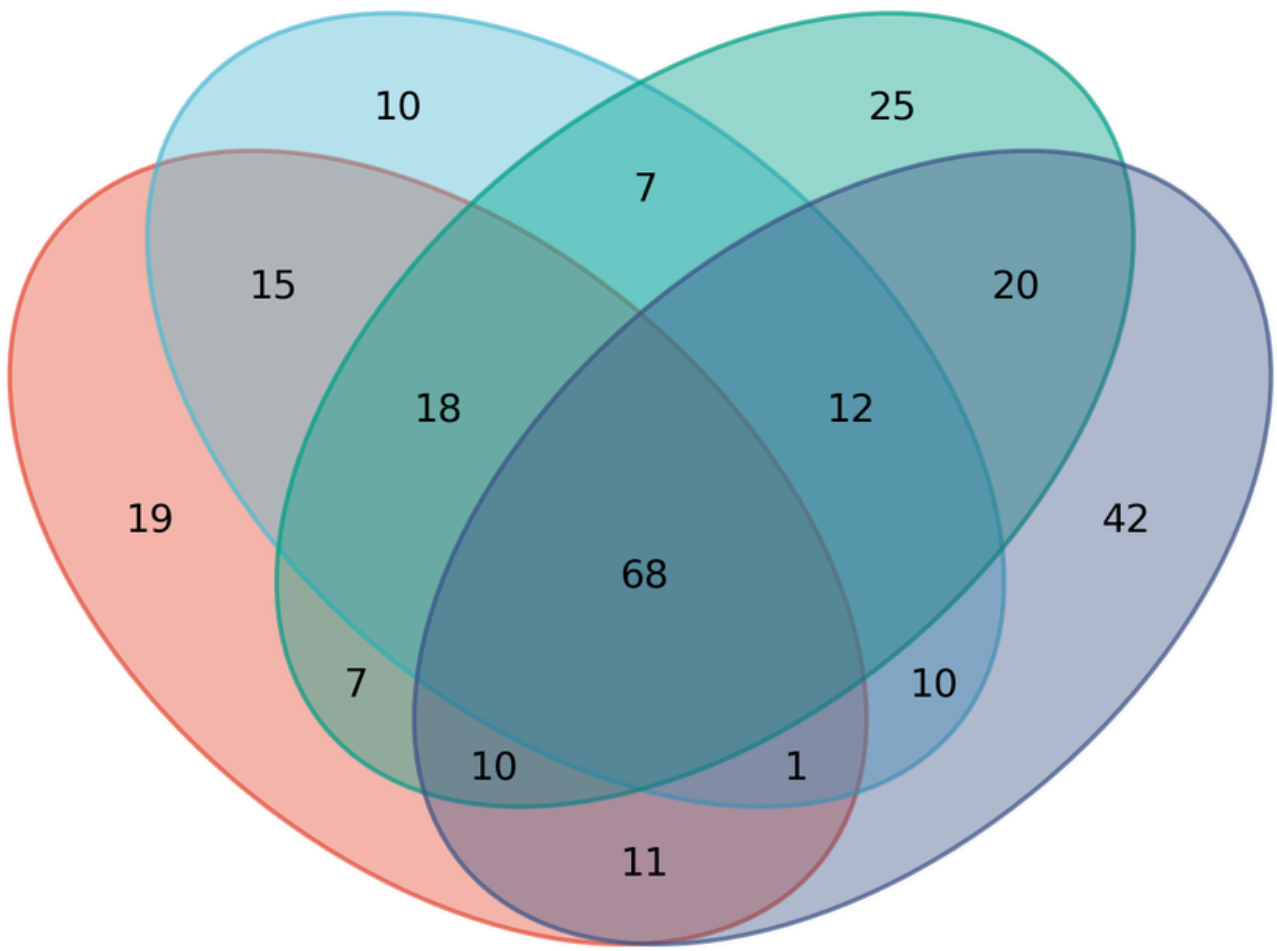
Pathway / GO enrichment of the pairwise memory-gene overlaps ($-\log_{10}$ FDR; dashed = 0.05; dot $\propto k$)



TF motif enrichment for Annotated Genes

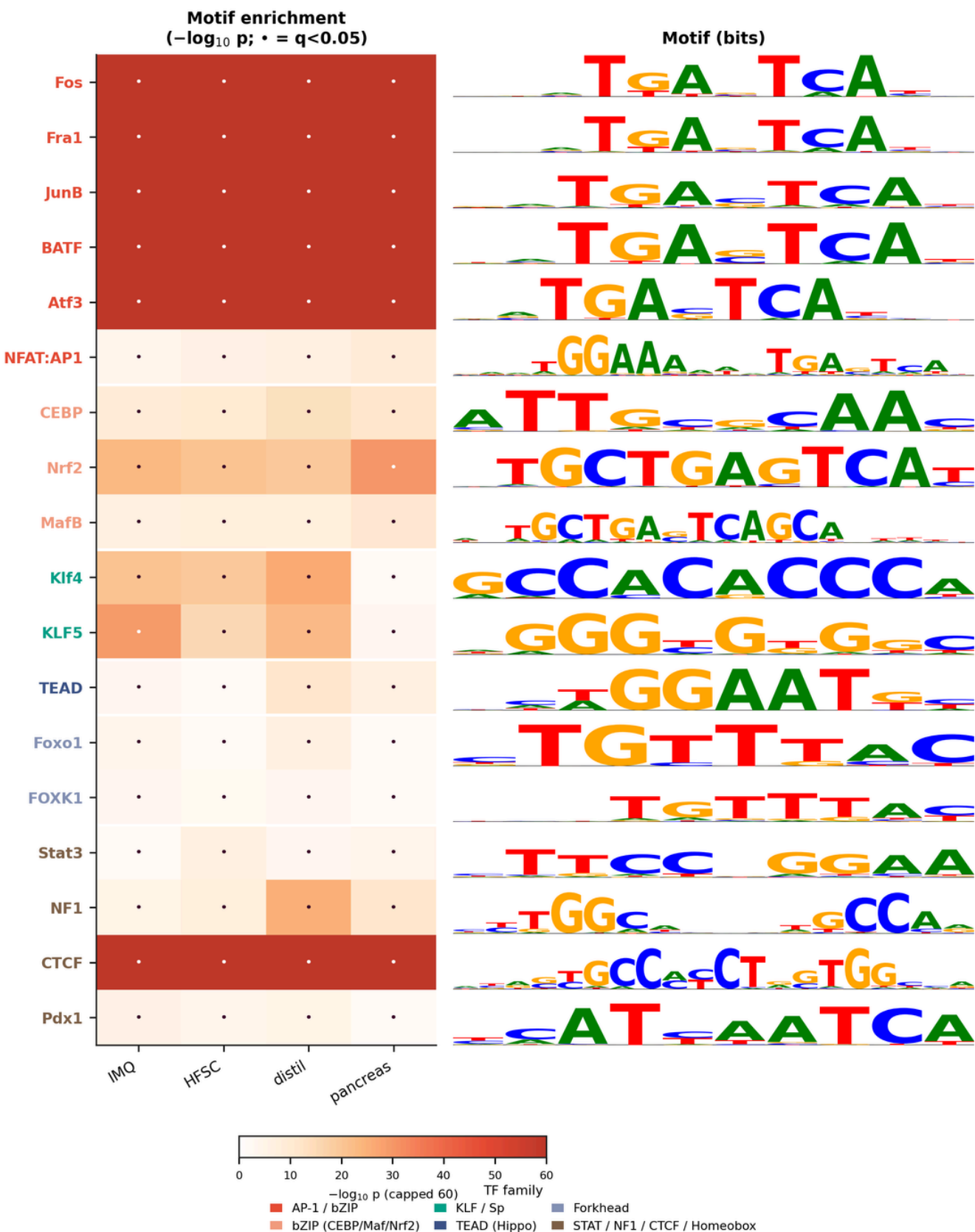
Enriched-TF overlap across four datasets (q<0.05)

- IMQ
- HFSC
- distil
- pancreas



four-way core = 68 TFs

Conserved TF-motif enrichment in memory peaks across four datasets



Inference

- Conducted a meta-analysis of Epigenetic memory across 4 different studies using 4 different stressors on 4 different in-vivo cell types.(all in mice)
- Exact Genomic coordinates **don't overlap** for the Memory Peaks (cell type specific encoding of memory)
- Significant overlap on the genes which are regulated by these memory peaks(Functional readout - **Elastin(ECM),Cell junction Remodeling.**
(**Common functional convergence of the memory**)
- **AP1 Family** as the universal regulator of epigenetic injury/wound memory across different cell types.

Future Prospect

Article | [Open access](#) | Published: 20 March 2026

AP-1 mediates cellular adaptation and memory formation

[Jingxin Li](#), [Pavithran T. Ravindran](#), [Aoife O'Farrell](#), [Gianna T. Busch](#), [Ryan H. Boe](#), [Zijian Niu](#), [Sean Woo](#), [Margaret C. Dunagin](#), [Naveen Jain](#), [Yogesh Goyal](#), [Kavitha Sarma](#), [Meenhard Herlyn](#) & [Arjun Raj](#) ✉

[Nature Communications](#) **17**, Article number: 4265 (2026) | [Cite this article](#)

13k Accesses | 12 Citations | 22 Altmetric | [Metrics](#)

Aging Cell Open Access **ANATOMICAL SOCIETY**

RESEARCH ARTICLE | [Open Access](#) |  

AP-1 is a regulatory transcription factor of inflammaging in the murine kidney and liver

[Xiaojie Yu](#), [Yuting Wang](#), [Yifan Song](#), [Xianda Gao](#), [Hongkui Deng](#) ✉

First published: 08 May 2023 | <https://doi.org/10.1111/accel.13858> | [VIEW METRICS](#)

Chronic inflammation and Cancer
Check the status of our common
Memory Peaks in pathological cases

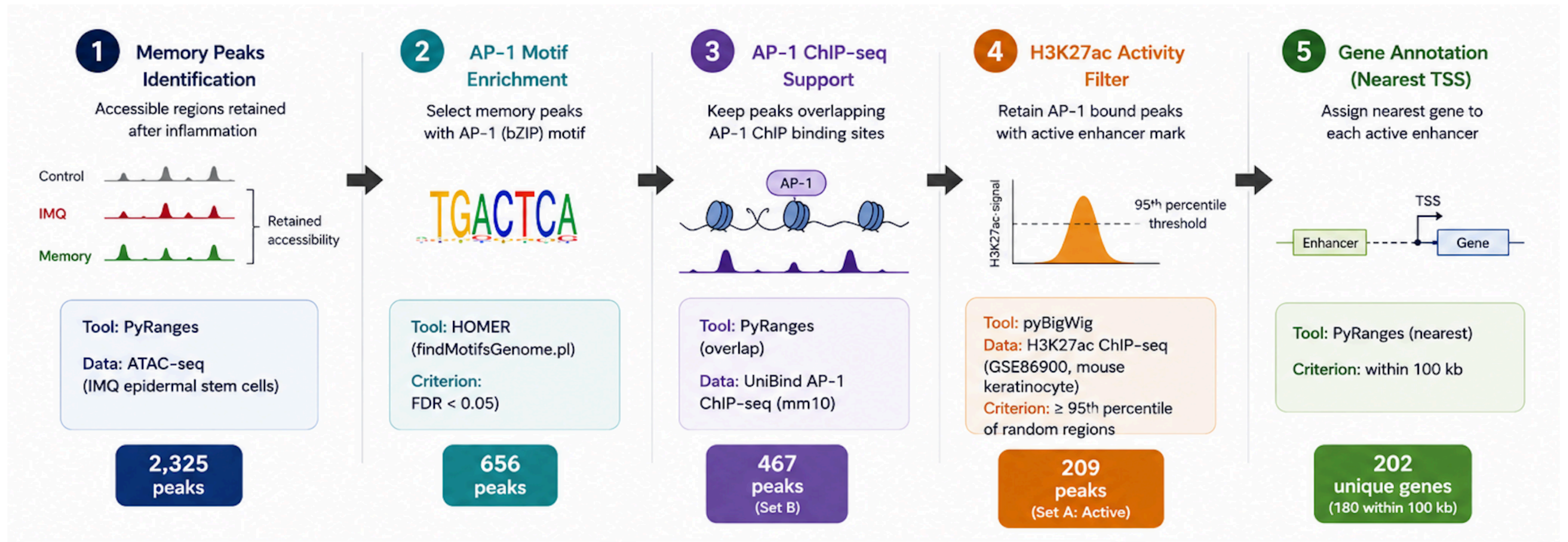
**Epigenetic Memory and
inflammaging**
See how these peaks Evolve as we
age.

Prospective Biomarker ?

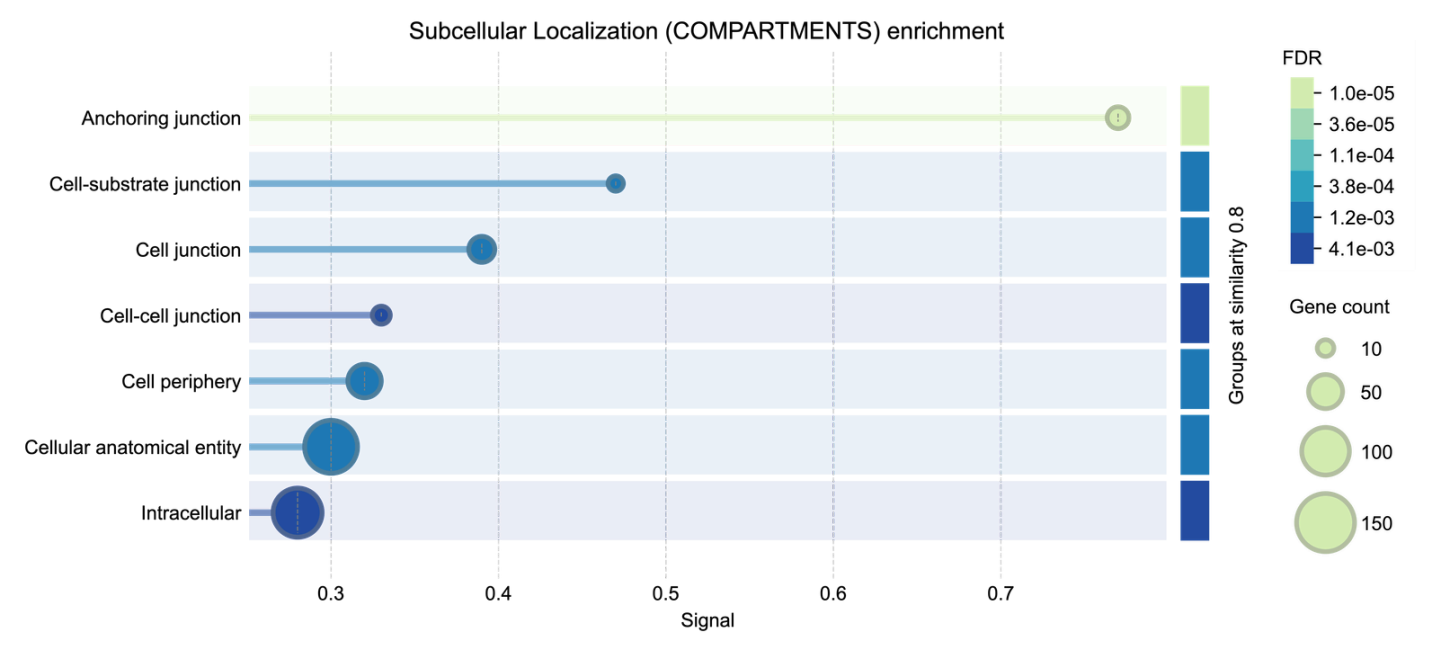
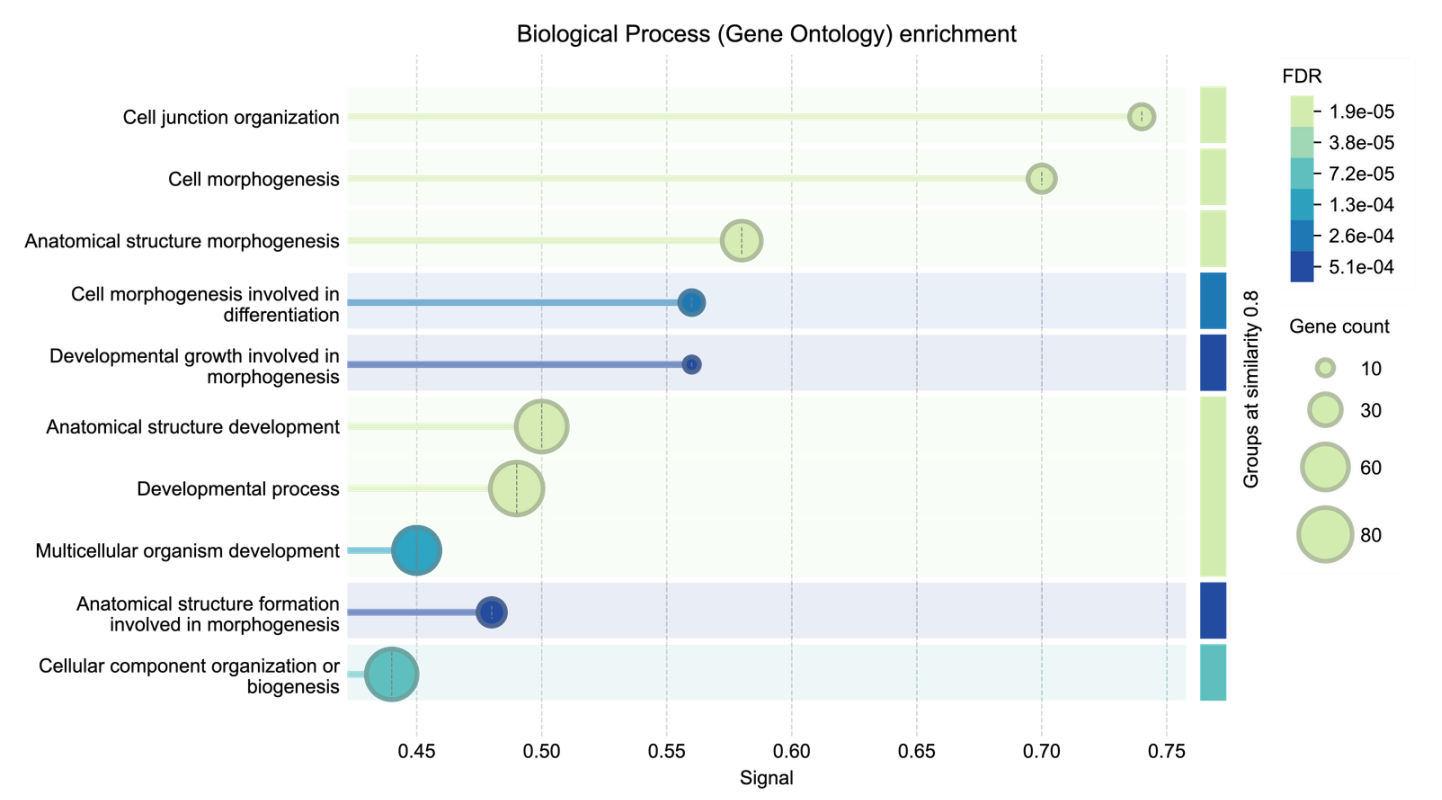
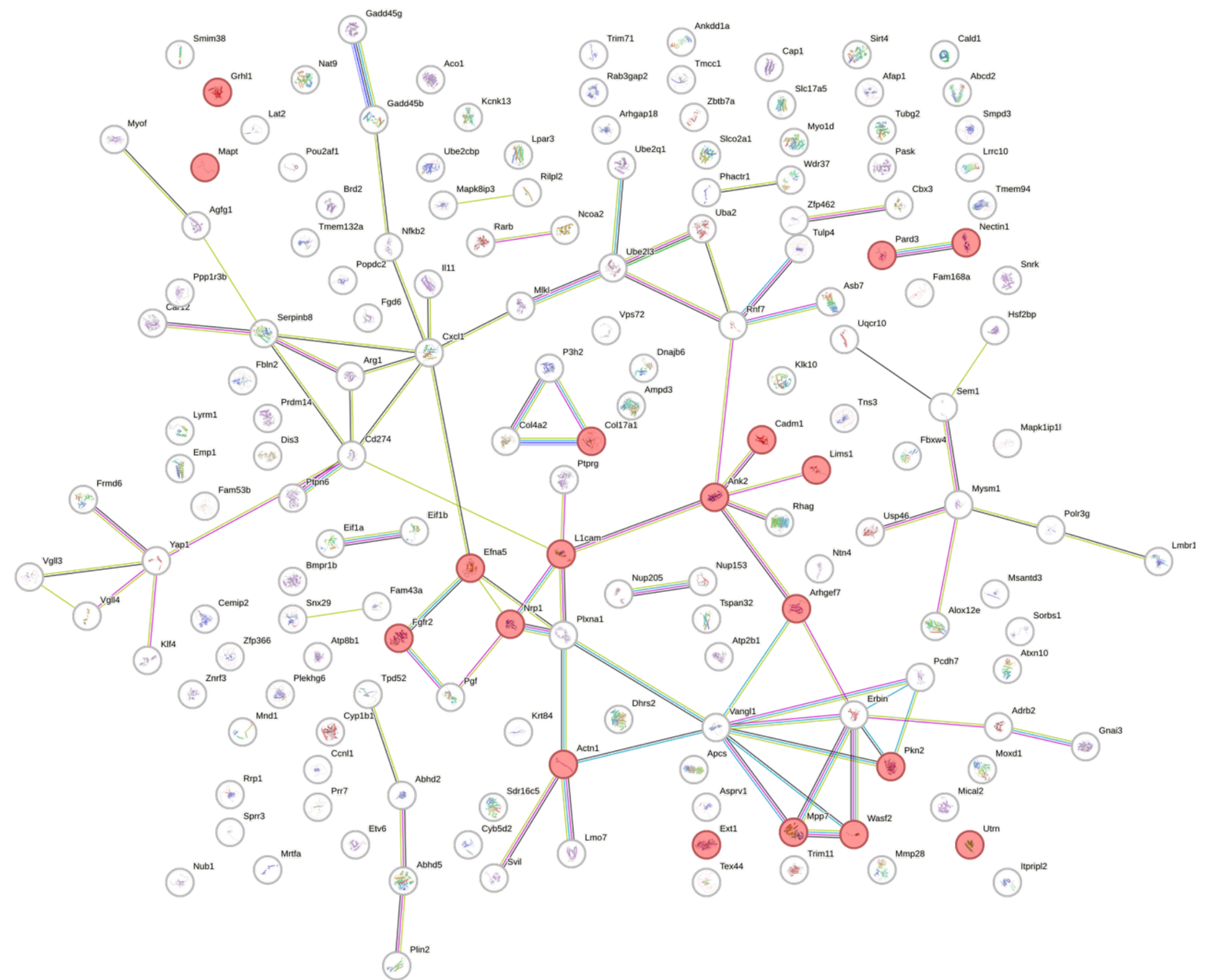
From memory peaks to **Bonafide AP1 regulated peaks**

Cell Stem Cell

Establishment, maintenance, and recall of inflammatory memory



String annotation of our gene

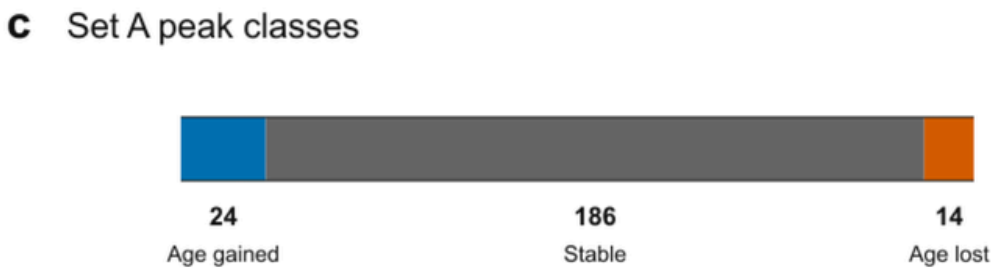
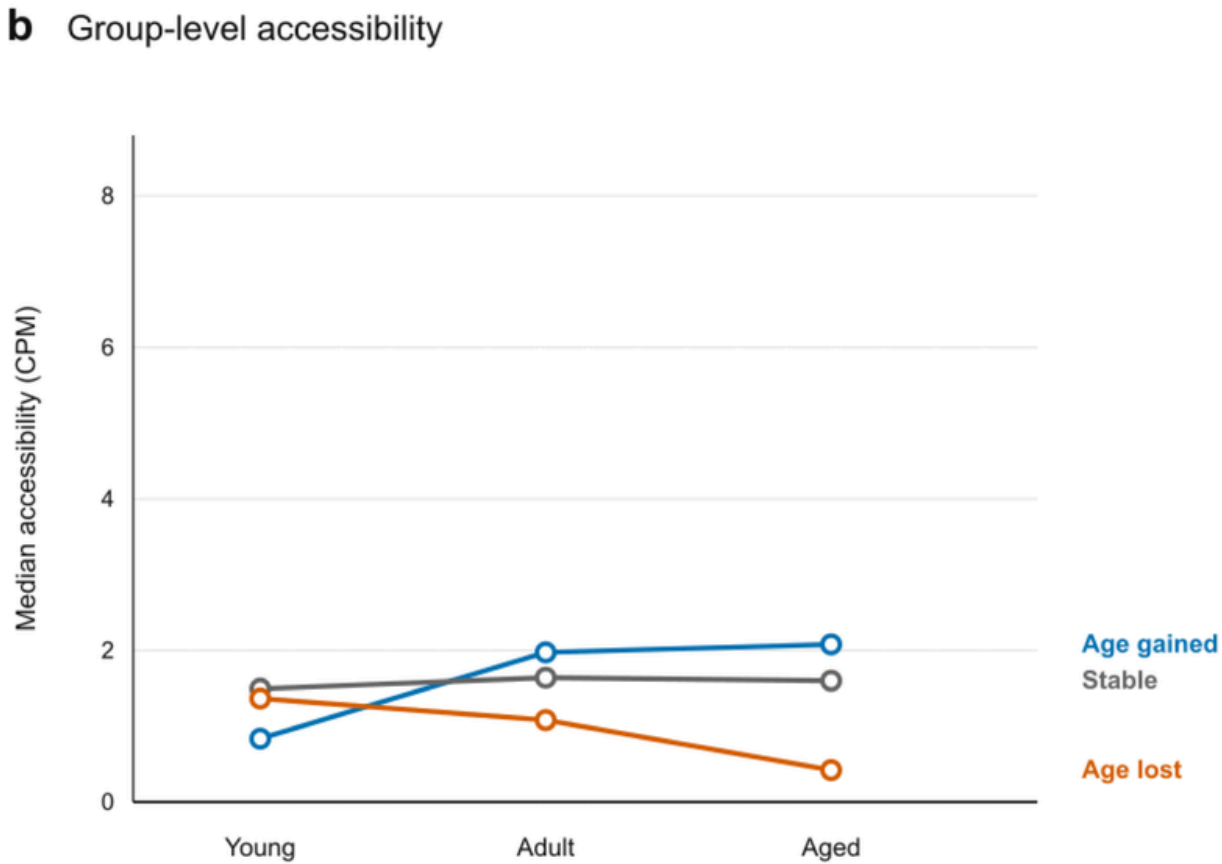
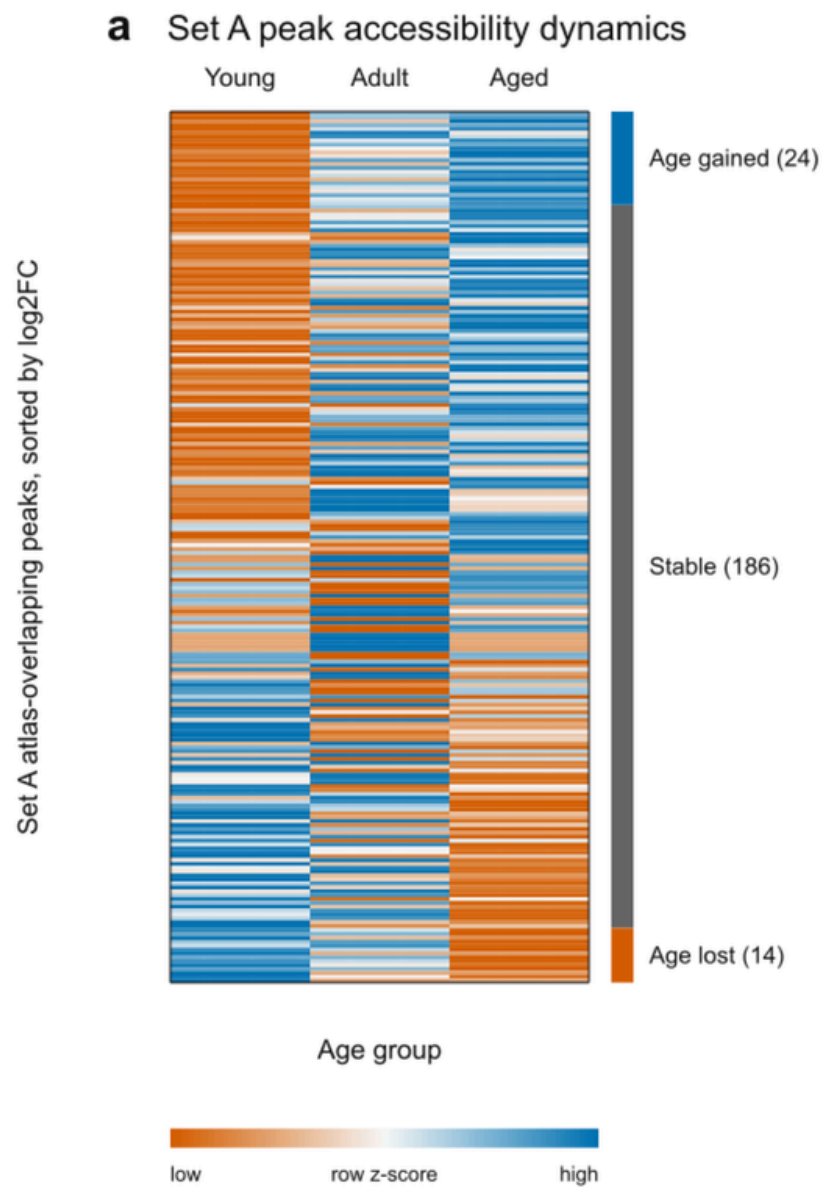
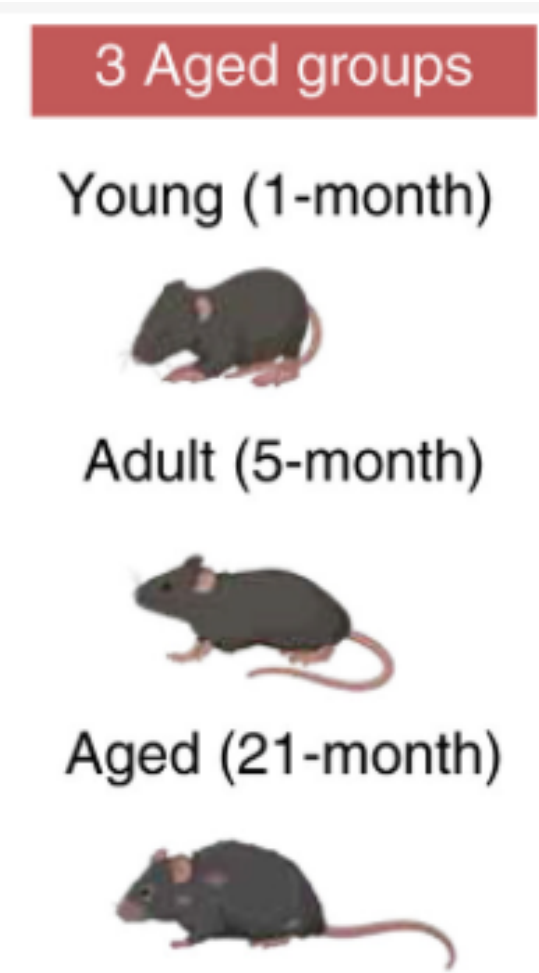


Age-dependent dynamics of AP-1-primed active memory enhancers

Organism-wide cellular dynamics and epigenomic re-modeling in mammalian aging

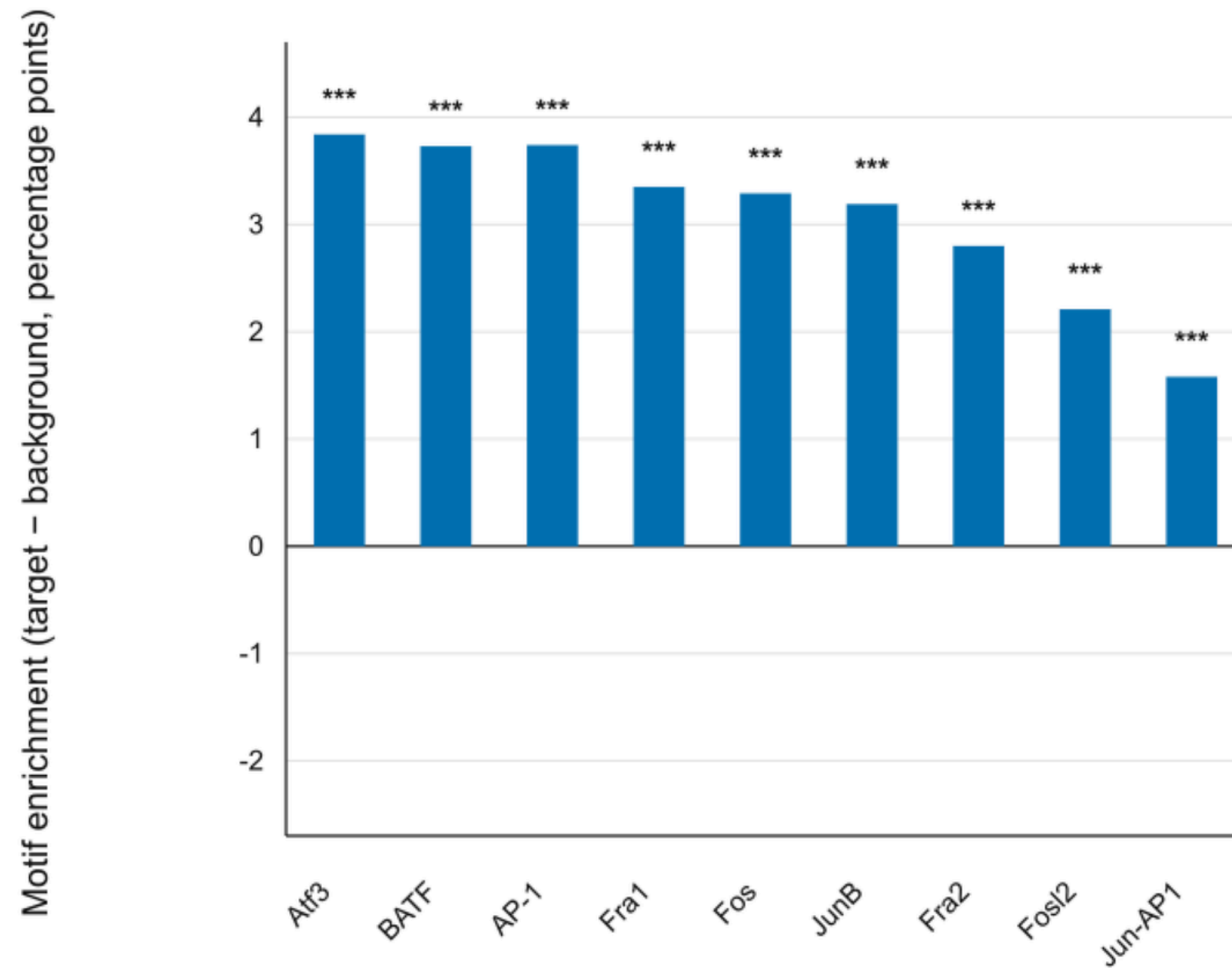
ZIYU LU , ZEHAO ZHANG , ZIHAN XU , ABDULRAOUF ABDULRAOUF, WEI ZHOU , AND JUNYUE CAO  [Authors Info & Affiliations](#)

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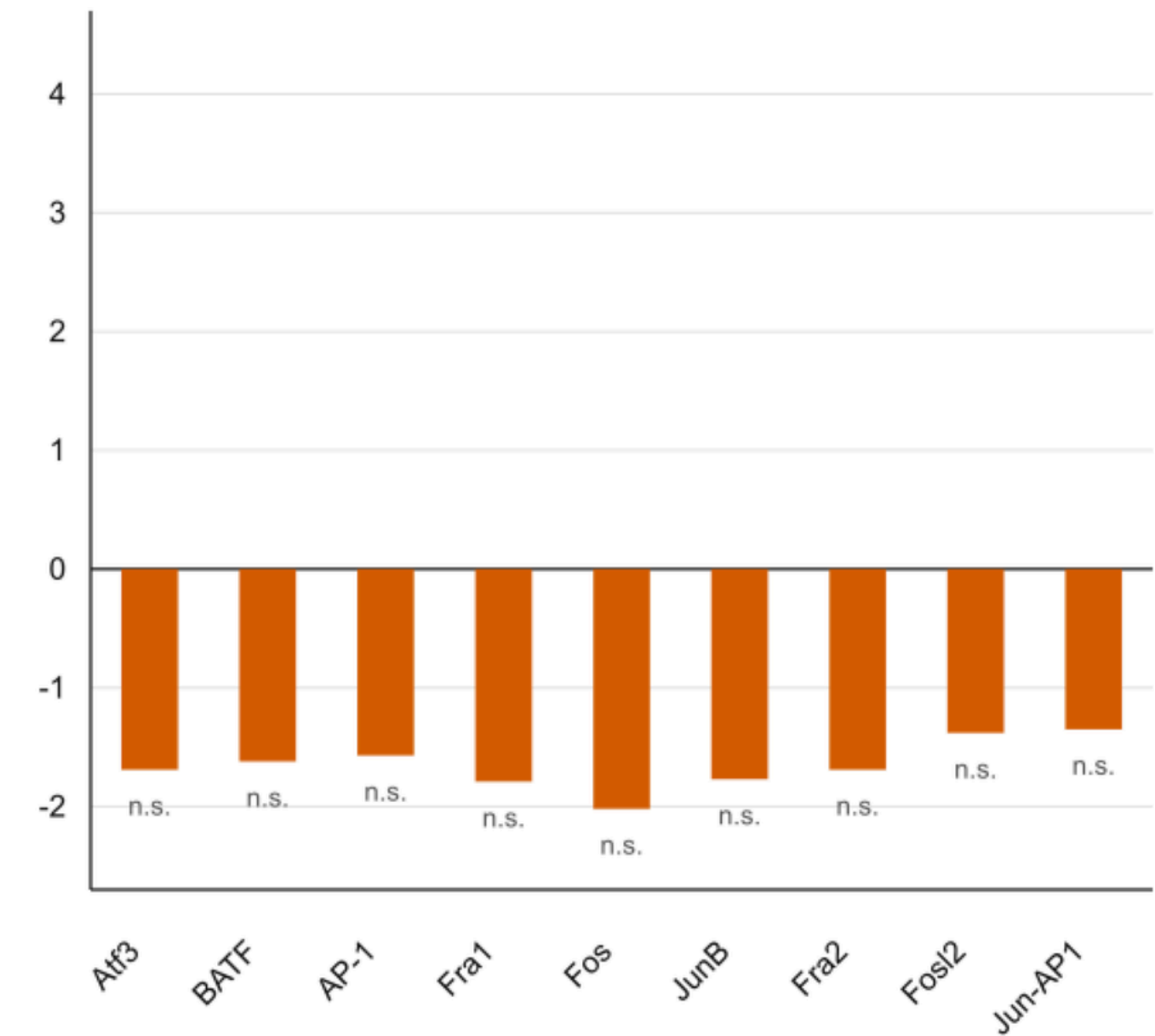


AP1 enrichment for DARs in skin epidermal cells

a Aged vs young



b Aged vs adult



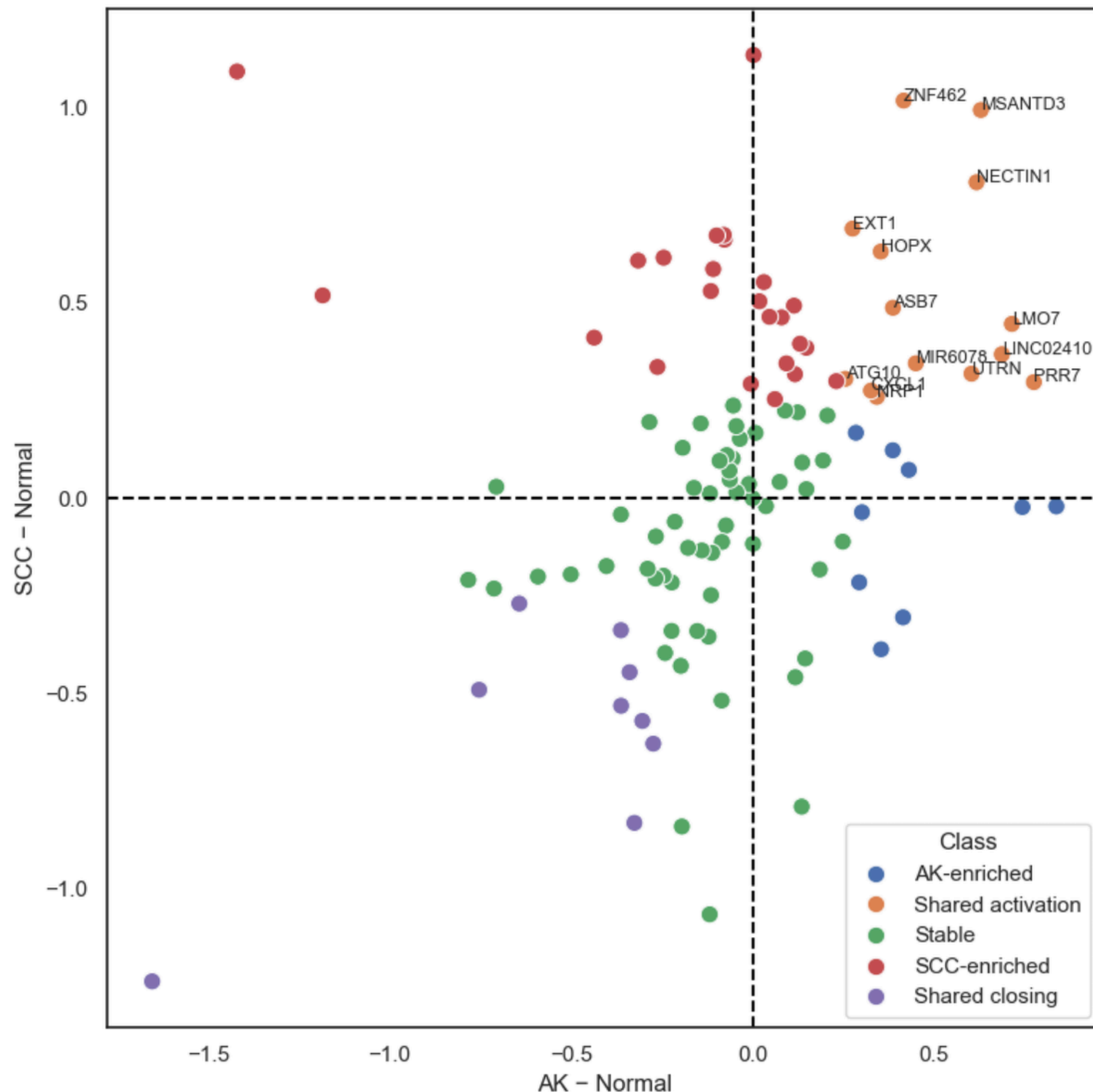
* $q < 0.05$ ** $q < 0.01$ *** $q < 0.001$; n.s., not significant

AP1 enrichment for Pre-Cancer and Squamous cell carcinoma

Pre cancer : Actinic keratosis
Cancer: Squamous cell carcinoma

14 genes primed in pre- cancer
and cancer

Strong association with skin cancer



> [Oncotarget](#). 2022 Oct 20;13:1166-1173. doi: 10.18632/oncotarget.28299.

Nectin-4 is widely expressed in head and neck squamous cell carcinoma

> [Eur J Cardiothorac Surg](#). 2014 Mar;45(3):514-20. doi: 10.1093/ejcts/ezt380. Epub 2013 Aug 16.

Photobiology

CXCL1 Inhibition Regulates UVB-Induced Skin Inflammation and Tumorigenesis in *Xpa*-Deficient Mice

Neuropilin-1 expression in squamous cell carcinoma of the oesophagus

> [PLoS One](#). 2017 Feb 17;12(2):e0171265. doi: [10.1371/journal.pone.0171265](#)

Recurrent rearrangements of the Myb/SANT-like DNA-binding domain containing 3 gene (*MSANTD3*) in salivary gland acinic cell carcinoma

Original Article

Carcinogenesis/Tumorigenesis

HOPX Exhibits Oncogenic Activity during Squamous Skin Carcinogenesis

HNSC TCGA cohort and survival analysis for our prognostic gene sets

